
Systematic Hazard Sampling: Minimal-Variance Inference for Discrete Diffusion and Flow Models

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Abstract

Uniform-noise discrete diffusion and flow models generate sequences non-autoregressively through iterative, context-dependent token replacements. However, these models are typically formulated as time-inhomogeneous continuous- or discrete-time Markov chains (CTMC/DTMC), sampled using independent Bernoulli change decisions per discretization step. This induces Poisson-binomial variance in per-position jump counts that grows with the number of required edits, leading to the common under-editing (residual noise) and over-editing (cascading substitutions) failure modes that degrade sample quality. We identify this sampler-induced variance as an orthogonal source of degradation, distinct from model-side errors and addressable purely at inference time. We propose *Systematic Hazard Sampling* (SHS), a training-free, drop-in, and hyperparameter-free inference principle for any sampler that admits a stay-vs.-replace decomposition. SHS models per-token edits as events driven by cumulative hazard (CTMC) or jump mass (DTMC) and triggers an edit whenever this quantity exceeds unit-spaced thresholds with a single random phase per position. For any fixed cumulative mass, this preserves the expected jump count while achieving the minimum conditional variance possible among unbiased integer estimators (at most $1/4$), without altering per-jump destination sampling. Experiments on four uniform-noise discrete diffusion and flow language models spanning $\sim 110\text{M}$ to $\sim 3\text{B}$ parameters show that SHS consistently improves sample quality across numbers of function evaluations (NFE). Code is available at <https://github.com/Jang-seunghwan/Systematic-Hazard-Sampling>.

1 Introduction

Non-autoregressive generation for high-dimensional discrete data (e.g., text, code) is a promising approach that can significantly reduce inference latency by updating all tokens in parallel. A growing family of methods recasts discrete generation as a time-inhomogeneous Markov process (CTMC or DTMC) that transports an easy prior p_0 to the data distribution p_1 , including discrete diffusion models (D3PM [Austin et al., 2021], SEDD [Lou et al., 2024], UDLM [Schiff et al., 2025]) and discrete flow matching (DFM) [Gat et al., 2024]. However, in uniform-noise-start models, meaningful generation typically requires multiple context-dependent edits per position.

This multi-jump flexibility comes at a practical cost in the standard step-based sampler. At each discretization step, every position independently decides whether to jump via a Bernoulli (or categorical) draw. The total number of edits per position is therefore a Poisson-binomial random variable whose variance $\sum_k p_{ik}(1 - p_{ik})$ can grow linearly with the cumulative jump mass. This sampler-

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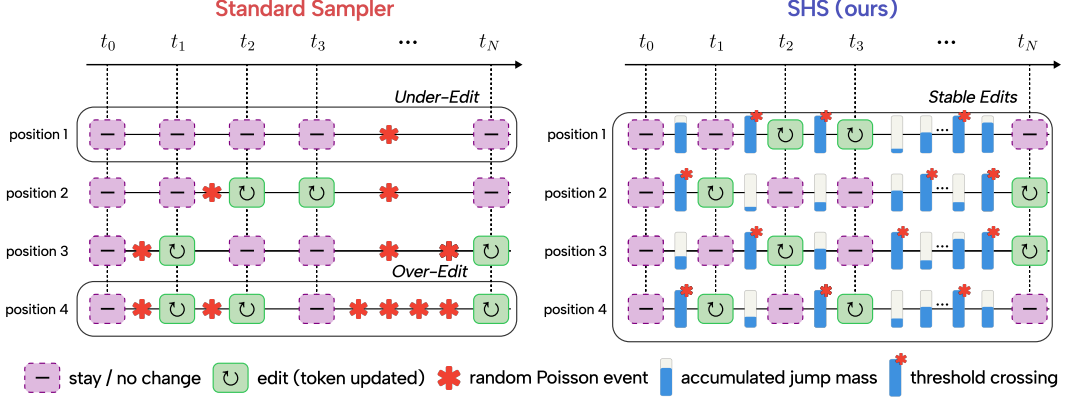


Figure 1: Standard step-based sampling (left) makes independent per-step Bernoulli edit decisions, leading to *under-edit* (position 1) and *over-edit* (position 4) failure modes. SHS (right) accumulates jump mass across steps and triggers edits at unit-spaced thresholds with a single random phase per position, producing stable, low-variance edit trajectories.

induced variance produces two characteristic failure modes at the tails of the jump-count distribution. *Under-editing* (too few jumps) occurs when insufficient substitutions leave residual noise or local inconsistencies, preventing full integration of contextual information before the process terminates. Conversely, *over-editing* (too many jumps) occurs when excessive substitutions cascade, disrupting even coherent segments and producing repetitions or distortions. Crucially, refining the time grid does not eliminate this variance: as the per-step Bernoulli probabilities shrink, $\sum_k p_{ik}(1 - p_{ik})$ approaches its Poisson limit S (the cumulative jump mass), so the variance *grows* with the number of function evaluations (NFE) rather than vanishing in the small-step limit.

These failure modes are not inherent to the uniform-noise formulation itself but are artifacts of the independent per-step sampling mechanism. Absorbing-state (mask-start) methods [Austin et al., 2021, Campbell et al., 2022]—where each position unmask exactly once—yield simple, low-variance trajectories, but reduce generation to any-order parallel unmasking without iterative refinement. Uniform-noise initialization [Austin et al., 2021, Lou et al., 2024, Schiff et al., 2025] allows each position to undergo multiple context-dependent jumps, providing genuine self-correction capability at the cost of increased trajectory complexity. Because existing step-based samplers for uniform-noise models typically realize replacements through independent per-step jumps, the resulting jump count and timing can be vulnerable to this sampler-induced variance, or “sampler luck”.

The goal of this paper is to preserve the expressive multi-jump dynamics of uniform-noise-start discrete generative models while suppressing this unnecessary sampler variance. To this end, we propose *Systematic Hazard Sampling (SHS)*, which requires no retraining, no additional hyperparameters, and no architectural modification—only a change to the inference-time sampling rule. SHS uses the cumulative hazard $S(t) = \int_0^t \lambda(s) ds$ in CTMC samplers (or the corresponding cumulative jump mass in DTMC samplers) to systematically place events in this space. SHS triggers jumps when the cumulative hazard $S(t)$ crosses integer boundaries offset by a random $\theta \sim \text{Uniform}(0, 1)$. Unlike prior works, which use probabilistic coin flips at fixed time steps to trigger the jumps, SHS preserves the expected jump count for any fixed cumulative hazard/jump mass while bounding the jump count variance to a theoretical minimum (at most $1/4$, *uniformly in NFE*). Moreover, unlike top- k /nucleus filtering, which biases the model’s learned destination distribution at inference time, SHS leaves the per-jump destination kernel $q_{i,k}$ unchanged and modifies only the temporal coupling of the stay-vs.-replace events. Consequently, SHS mitigates the long-tail waiting times and “sampler luck” that are especially pronounced in uniform-noise starts; this simultaneously suppresses under- and over-editing.

Empirically, we evaluate SHS on four uniform-noise discrete diffusion and flow models spanning $\sim 110\text{M}$ to $\sim 3\text{B}$ parameters (UDLM, GIDD, DFM, and a 3B uniform diffusion model). SHS consistently improves generative perplexity (Gen. PPL) across NFE budgets; the predicted jump-count concentration holds in the real sampler, and under-editing turns out to be the costly side of count deviation (Sections 4.2–4.3).

2 Preliminaries

Let $\mathcal{V}$ be a vocabulary, N the sequence length, n the number of discretization steps, and $x = (x_1, \dots, x_N) \in \mathcal{V}^N$ a sequence. Discrete diffusion and flow language models define time-inhomogeneous stochastic processes that transport a simple source distribution p_0 to the data distribution p_1 . The source and transition family determine whether generation is essentially single-edit or multi-edit at each position.

Masked/absorbing vs. uniform-noise starts. Absorbing-state or mask-start models augment the vocabulary with a special state [MASK] and generate by resolving masked positions into vocabulary tokens. In the standard unmask-only sampler, once a position leaves [MASK], it is fixed:

$$p_{ik}(x) = 0 \quad \text{whenever } x_i \in \mathcal{V}, \quad (1)$$

where $p_{ik}(x)$ is the probability that position i changes at step k . Thus each position can edit at most once: with J_i denoting the total jump count at position i over the trajectory, $J_i \in \{0, 1\}$ almost surely (see Appendix D.4 for a formal statement). This monotone structure is simple and stable, and has led to strong masked/absorbing diffusion language models (DLMs) [Austin et al., 2021, Sahoo et al., 2024, Shi et al., 2024, Ou et al., 2025]. Recent variants of mask-start models restore revision ability by adding remasking or correction modules [Wang et al., 2025, Schiff et al., 2026]. In contrast, uniform-noise models [Schiff et al., 2025, von Rütte et al., 2025] initialize positions from vocabulary-valued noise and allow tokens to be replaced repeatedly, giving the sampler a genuine multi-edit trajectory; recent scaling studies of uniform and interpolating diffusion at billion-parameter scale [von Rütte et al., 2026, Sahoo et al., 2026] make this regime practically important. Discrete flow matching provides a related probability-path view for categorical generation [Gat et al., 2024, Monsefi et al., 2026]. All of these directions improve the trained model itself—its noise structure, training objective, or learned distribution—but leave an orthogonal source of degradation, sampler-induced variance in step-based simulation, largely unaddressed.

A stay-vs.-replace view of CTMC/DTMC samplers. Discrete generative models differ in how they define or learn the reverse dynamics: structured DTMC kernels [Hoogeboom et al., 2021, Austin et al., 2021], CTMC reverse processes [Campbell et al., 2022], score or ratio parameterizations [Lou et al., 2024], uniform/hybrid marginal-transition constructions [Schiff et al., 2025, von Rütte et al., 2025], and discrete flow-matching probability paths [Gat et al., 2024, Monsefi et al., 2026]. We do not require these formulations to be identical. We only require that the *sampler* admit a local stay-vs.-replace decomposition.

CTMC form. A time-inhomogeneous CTMC on $\mathcal{V}^N$ is specified by a generator Q_t . For each position i , write $\lambda_i(t, x) \geq 0$ for the escape rate and $q_{i,t}(\cdot | x) \in \Delta(\mathcal{V} \setminus \{x_i\})$ for the destination distribution (probability simplex) conditional on an edit. With $x^{(i \leftarrow v)}$ denoting the sequence obtained by replacing x_i with v , the local off-diagonal rates can be written as

$$Q_t(x^{(i \leftarrow v)} | x) = \lambda_i(t, x) q_{i,t}(v | x), \quad v \neq x_i, \quad (2)$$

and $Q_t(x | x) = -\sum_i \lambda_i(t, x)$, the negative total escape rate, so that each row of the generator sums to zero [Norris, 1997]. In an Euler or τ -leaping discretization on grid points $t_k = kh$ with step size $h = 1/n$, the model supplies a per-step change mass $p_{ik} \in [0, 1]$, for example $p_{ik} = 1 - \exp(-h\lambda_i(t_k, x))$ under a frozen-rate Poisson clock, or $p_{ik} \approx h\lambda_i(t_k, x)$ for small steps.

DTMC form. For a discrete-time reverse kernel, let $P_{i,k}(\cdot | x) \in \Delta(\mathcal{V})$ be the per-position categorical update distribution. Any such kernel can be decomposed as

$$p_{ik}(x) = 1 - P_{i,k}(x_i | x), \quad q_{i,k}(v | x) = \frac{P_{i,k}(v | x)}{p_{ik}(x)} \quad (v \neq x_i), \quad (3)$$

where $p_{ik}(x)$ is the probability that the position changes and $q_{i,k}$ is the destination distribution conditional on changing. If $p_{ik}(x) = 0$, the destination distribution is irrelevant.

Unified event view. Both CTMC τ -leaping and DTMC categorical updates can be written using a per-step edit indicator $B_{ik} \in \{0, 1\}$ and a replacement token $V_{ik} \in \mathcal{V}$ as

$$B_{ik} \sim \text{Bernoulli}(p_{ik}), \quad X_{t_{k+1}}^i = \begin{cases} X_{t_k}^i, & B_{ik} = 0, \\ V_{ik}, & B_{ik} = 1, \end{cases} \quad V_{ik} \sim q_{i,k}(\cdot | X_k), \quad (4)$$

Define the cumulative hazard or jump mass

$$S_{i,k} = \sum_{j=0}^{k-1} p_{ij}, \quad S_{i,n} = \sum_{j=0}^{n-1} p_{ij} \approx \int_0^1 \lambda_i(t, X_t) dt. \quad (5)$$

The variance statements below are conditional on a prescribed or realized sequence of masses $\{p_{ik}\}_{k=0}^{n-1}$; in the full sampler, these masses are state-dependent.

Jump-count variance under grid refinement. Conditional on a fixed mass sequence $\{p_{ik}\}_{k=0}^{n-1}$, the standard sampler draws independent Bernoulli indicators. The per-position jump count

$$J_i = \sum_{k=0}^{n-1} B_{ik}$$

therefore follows a Poisson-binomial law with

$$\mathbb{E}[J_i \mid \{p_{ik}\}] = S_{i,n}, \quad \text{Var}(J_i \mid \{p_{ik}\}) = \sum_{k=0}^{n-1} p_{ik}(1 - p_{ik}). \quad (6)$$

Refining the time grid does not remove this conditional jump-count variance. Fixing a single position and dropping the position index i for clarity (the index returns in Section 3), for a fixed cumulative mass $S = \sum_{k=1}^n p_k$,

$$\text{Var}(J \mid \{p_k\}) = \sum_{k=1}^n p_k(1 - p_k) = S - \sum_{k=1}^n p_k^2. \quad (7)$$

If $\max_k p_k \rightarrow 0$ while $\sum_k p_k = S$, then

$$\sum_{k=1}^n p_k^2 \leq \left(\max_k p_k\right) \sum_{k=1}^n p_k = \left(\max_k p_k\right) S \rightarrow 0, \quad \text{Var}(J \mid \{p_k\}) \rightarrow S. \quad (8)$$

Thus finer discretization may reduce numerical integration error, but it does not eliminate the independent Bernoulli jump-count fluctuations; in the small-step limit, the count variance approaches the Poisson limit associated with cumulative hazard S .¹

3 Systematic Hazard Sampling

We propose *Systematic Hazard Sampling (SHS)*, adapting systematic event simulation to the *step-based* samplers used in CTMC/DTMC discrete generative models. SHS replaces the independent per-step change decisions $B_{ik} \sim \text{Bernoulli}(p_{ik})$ (Section 2) with *single-phase* systematic placement in cumulative hazard/jump-mass space. Concretely, SHS maintains the cumulative mass $S_{i,k}$ from (5) and triggers a jump whenever $S_{i,k}$ crosses unit-spaced thresholds $\{\theta_i + m\}_{m \in \mathbb{Z}_{\geq 0}}$ for a single random phase $\theta_i \sim \text{Uniform}(0, 1)$. The complete pseudocode of SHS is shown in Algorithm 1.

One boundary per step. Since $p_{ik} \leq 1$ holds by definition for DTMC kernels and is a prerequisite for valid CTMC τ -leaping, at most one unit boundary can be crossed per step, so the if statement on Line 14 is sufficient. Otherwise, Line 14 can be replaced by a while-loop that triggers all crossed boundaries.

SHS changes only the *timing* of jump events, by coupling the Bernoulli decisions across steps through the single phase θ_i ; Appendix D.2 quantifies the resulting shift in edit timing. The destination sampling $q_{i,k}$ is unchanged: whenever an edit is triggered, the replacement token is drawn from the same destination kernel as in the standard sampler.

¹The common uniform refinement $p_k = S/n$ satisfies the condition exactly. A frozen-rate CTMC step with $p_k = 1 - \exp(-S/n)$ also satisfies $\max_k p_k \rightarrow 0$ and $\sum_k p_k \rightarrow S$, yielding the same limit.

Jump-count concentration. Let $S_i^{\text{tot}} := S_{i,n} = \sum_{k=0}^{n-1} p_{ik}$ denote the total cumulative hazard/jump mass accumulated at position i . Write $S_i^{\text{tot}} = I_i + f_i$ with $I_i = \lfloor S_i^{\text{tot}} \rfloor$ and $f_i \in [0, 1)$. Conditioning on a fixed realized S_i^{tot} , SHS counts how many thresholds $\theta_i + m$ are crossed by S_i^{tot} , so we have

$$J_i = I_i + \mathbf{1}[\theta_i < f_i], \quad (9)$$

so $J_i \in \{I_i, I_i + 1\}$ conditional on the realized cumulative mass S_i^{tot} , and the conditional long-tail jump-count fluctuations of step-based Bernoulli sampling are eliminated for that fixed cumulative mass S_i^{tot} . In the full model the masses $p_{ik}(x)$ are state-dependent, so S_i^{tot} itself is a random variable whose distribution may differ between SHS and standard sampling (see Appendix D.3 for discussion). As a result, SHS suppresses the characteristic under-edit (too few substitutions) and over-edit (cascading substitutions) failure modes that arise under uniform-noise initialization.

Algorithm 1 Systematic Hazard Sampling (SHS) for CTMC/DTMC discrete generative models

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1: Input:
2:   - Initial state  $X \sim p_0$ , Time grid  $t_k = kh$  for  $k = 0, \dots, n$ 
3:   - A routine  $\text{MODELSTEP}(t_k, X_{t_k}, i)$  that returns: (cf. (2), (3))
4:     - change mass  $p_{ik} \in [0, 1]$ 
5:     - destination  $q_{i,k}(\cdot \mid X_{t_k})$ 
6: Output: Final sample  $X$  at  $t_n$  (i.e.,  $t = 1$ )
7: for  $i = 1$  to  $N$  do
8:    $S_i \leftarrow 0$ ;  $m_i \leftarrow 0$ ; Draw  $\theta_i \sim \text{Uniform}(0, 1)$ 
9: end for
10: for  $k = 0, 1, \dots, n - 1$  do
11:   for  $i = 1$  to  $N$  do
12:      $(p_{ik}, q_{i,k}) \leftarrow \text{MODELSTEP}(t_k, X_{t_k}, i)$ 
13:      $S_i \leftarrow S_i + p_{ik}$ 
14:     if  $S_i \geq \theta_i + m_i$  then  $\{/*crossed the next hazard boundary*/\}$ 
15:        $X_{t_{k+1}}^i \sim q_{i,k}(\cdot \mid X_{t_k})$ ;  $m_i \leftarrow m_i + 1$ 
16:     else
17:        $X_{t_{k+1}}^i \leftarrow X_{t_k}^i$ 
18:     end if
19:   end for
20: end for
21: return  $X_{t_n}$ 

```

3.1 Theoretical Properties

SHS is built on a simple primitive: *randomized rounding* of a nonnegative real mass S into an integer count $J \in \mathbb{Z}_{\geq 0}$ using a *single* uniform random variable. Writing $S = I + f$ with $I := \lfloor S \rfloor$ and $f := S - I \in [0, 1)$ (a decomposition we use throughout), the rounding rule is $J = I + \mathbf{1}[\theta < f]$ for $\theta \sim \text{Uniform}(0, 1)$. In SHS, this mass is the total cumulative hazard/jump mass S_i^{tot} at each position (9).

Proposition 1 (Jump-Count Unbiasedness). Fix $S \geq 0$ and draw $\theta \sim \text{Uniform}(0, 1)$. Write $S = I + f$ with $I := \lfloor S \rfloor$ and $f := S - I \in [0, 1)$. Define $J := I + \mathbf{1}[\theta < f]$. Then $\mathbb{E}[J] = S$.

The randomized rounding rule used by SHS produces an integer jump count whose expectation matches the cumulative mass exactly: for a given cumulative mass, SHS does not systematically over- or under-edit relative to the model's intended total.

Proposition 2 (Minimal Variance). With J as in Proposition 1, we have

$$\text{Var}(J) = f(1 - f) \leq \frac{1}{4}. \quad (10)$$

Moreover, among all $\mathbb{Z}_{\geq 0}$ -valued unbiased estimators of S , this variance is the minimum possible.

The realized jump count never deviates from the cumulative mass by more than one, and its variance is at most $1/4$ regardless of S or the discretization budget—the smallest variance any integer-valued unbiased rounding can achieve.

Proof sketches. Write $S = I + f$ with $I = \lfloor S \rfloor$ and $f = S - I \in [0, 1)$. Then $J = I + B$ where $B = \mathbf{1}[\theta < f] \sim \text{Bernoulli}(f)$. Hence $\mathbb{E}[J] = I + f = S$ and $\text{Var}(J) = f(1 - f) \leq 1/4$. Minimality holds because, for any integer-valued $\tilde{J}$ with $\mathbb{E}[\tilde{J}] = S$ and $U := \tilde{J} - I$, the identity $(U - f)^2 = U(U - 1) + (1 - 2f)U + f^2$ gives $\text{Var}(\tilde{J}) = \mathbb{E}[U(U - 1)] + f(1 - f) \geq f(1 - f)$, with equality if and only if $\tilde{J} \in \{I, I + 1\}$ almost surely. See Appendix B.2 for the full proof (and Appendix B.1 for Proposition 1).

Interpretation. For a fixed mass S , standard step-based simulation yields a Poisson-binomial count with variance $\sum_k p_k(1 - p_k)$, which can scale linearly with S in the many-edit regime. In contrast, SHS concentrates the jump count to the two adjacent integers ($\lfloor S \rfloor$ or $\lceil S \rceil$), eliminating long-tail “sampler luck” in the number of edits. This statement is about the binary jump-count process; the categorical destination kernel used at each realized edit is left unchanged. Section 4.2 measures this concentration in the real sampler, and Section 4.3 shows that the deviations it removes are the ones that cost quality.

3.2 Reducing Event Variance without Collapsing Destination Diversity

A natural concern with Proposition 2 is whether reducing jump-count variance also reduces the multimodality that stochastic sampling provides. We separate three sources of randomness in a discrete language model sampler: (i) the initial source sample $X_0 \sim p_0$, which acts as the global generation seed; (ii) the binary edit schedule $B = (B_{ik})_{i \leq N, 0 \leq k < n} \in \{0, 1\}^{N \times n}$, which decides when each position is edited; and (iii) the categorical destination draws $V_{ik} \sim q_{i,k}(\cdot \mid X_{t_k})$, which determine what token is written when an edit occurs. SHS modifies only the second source: it changes the temporal coupling of the binary edit indicators while preserving the initial source distribution p_0 and the categorical destination kernels $q_{i,k}$. The following two structural facts make this asymmetry precise; proofs are in Appendix C.

Write $B_{ik}^{\text{std}}, J_i^{\text{std}}$ for the standard sampler and $B_{ik}^{\text{SHS}}, J_i^{\text{SHS}}$ for Algorithm 1.

Proposition 3 (Per-step event-marginal preservation). *For any fixed mass sequence $\{p_{ik}\}_{k=0}^{n-1}$ with $p_{ik} \leq 1$,*

$$\mathbb{P}_{\theta_i}(B_{ik}^{\text{SHS}} = 1 \mid \{p_{ij}\}_{j=0}^{n-1}) = p_{ik} = \mathbb{P}(B_{ik}^{\text{std}} = 1 \mid \{p_{ij}\}_{j=0}^{n-1}). \quad (11a)$$

$$\mathbb{E}_{\theta_i}[J_i^{\text{SHS}} \mid \{p_{ij}\}_{j=0}^{n-1}] = \sum_{k=0}^{n-1} p_{ik} = \mathbb{E}[J_i^{\text{std}} \mid \{p_{ij}\}_{j=0}^{n-1}]. \quad (11b)$$

where J_i^s is the total number of jump events at position i under sampler $s \in \{\text{SHS}, \text{std}\}$.

Proposition 3 states that at every step, the marginal probability of an edit under SHS equals that under the standard sampler of Section 2. SHS therefore preserves the model’s per-step edit budget along any fixed mass sequence and changes only the coupling between steps; Section 4.2 checks the same conditions on the masses the state-dependent sampler actually produces.

Now, given the full binary event pattern $B = (B_{ik})$, let π_s denote the induced law of B for sampler $s \in \{\text{std}, \text{SHS}\}$. Given an externally specified realization b of B , let $\mathcal{K}_b$ denote the trajectory law obtained by drawing $X_0 \sim p_0$ and applying the common destination rule $q_{i,k}(\cdot \mid X_{t_k})$ at exactly the events specified by b .

Proposition 4 (Destination-kernel identity under fixed events). *For any state x , position i , step k , and token $v \neq x_i$,*

$$\mathbb{P}^s(X_{t_{k+1}}^i = v \mid B_{ik} = 1, X_{t_k} = x) = q_{i,k}(v \mid x) \quad \text{for both } s \in \{\text{std}, \text{SHS}\}. \quad (12)$$

Consequently, both samplers induce the same trajectory kernel $\mathcal{K}_b$ for any fixed event pattern b .

Once an edit is triggered, both samplers draw the replacement token from the same destination distribution. SHS does not truncate, sharpen, or otherwise reshape what the model wants to write at any edit; it only changes when edits are scheduled.

Table 1: **Generative perplexity under NFE budgets** (Gen. PPL, lower is better; bold = better between Standard and SHS). Δ shows SHS improvement over Standard. $\pm$: standard deviation over seeds (5 for UDLM and GIDD, 3 for DFM; single seed for 3B).

UDLM						GIDD					
NFE	Standard		SHS		Δ	Standard		SHS		Δ	
	Entropy	Gen. PPL↓	Entropy	Gen. PPL↓		Entropy	Gen. PPL↓	Entropy	Gen. PPL↓		
4	6.633 ± 0.008	559.3 ± 6.3	6.586 ± 0.007	535.6 ± 5.6	-4.2%	7.479 ± 0.010	1696.2 ± 14.5	7.438 ± 0.010	1606.4 ± 13.6	-5.3%	
8	6.739 ± 0.007	336.3 ± 2.8	6.670 ± 0.011	316.4 ± 2.2	-5.9%	7.398 ± 0.005	528.9 ± 4.1	7.320 ± 0.009	455.6 ± 6.6	-13.9%	
16	6.806 ± 0.013	248.0 ± 2.7	6.742 ± 0.012	232.1 ± 3.7	-6.4%	7.341 ± 0.009	227.7 ± 3.7	7.261 ± 0.011	189.5 ± 1.9	-16.7%	
32	6.839 ± 0.017	208.9 ± 4.4	6.791 ± 0.002	198.2 ± 3.9	-5.1%	7.327 ± 0.009	150.0 ± 1.5	7.232 ± 0.009	119.3 ± 0.3	-20.5%	
64	6.860 ± 0.005	193.4 ± 1.7	6.810 ± 0.005	184.8 ± 3.2	-4.5%	7.331 ± 0.007	120.9 ± 1.8	7.233 ± 0.011	94.6 ± 0.9	-21.7%	
128	6.880 ± 0.010	188.9 ± 2.6	6.822 ± 0.017	175.7 ± 3.7	-7.0%	7.356 ± 0.010	111.7 ± 1.5	7.235 ± 0.007	85.1 ± 1.4	-23.8%	
256	6.869 ± 0.010	182.1 ± 2.7	6.815 ± 0.011	171.8 ± 3.2	-5.7%	7.371 ± 0.012	107.7 ± 1.8	7.245 ± 0.013	80.2 ± 0.9	-25.6%	
512	6.884 ± 0.013	183.4 ± 3.4	6.826 ± 0.009	169.0 ± 2.4	-7.9%	7.378 ± 0.006	103.9 ± 1.1	7.241 ± 0.011	77.4 ± 1.0	-25.5%	
1024	6.888 ± 0.012	181.6 ± 2.7	6.818 ± 0.011	167.2 ± 1.0	-7.9%	-	-	-	-	-	
3B uniform diffusion						DFM					
NFE	Standard		SHS		Δ	Standard		SHS		Δ	
	Entropy	Gen. PPL↓	Entropy	Gen. PPL↓		Entropy	Gen. PPL↓	Entropy	Gen. PPL↓		
4	6.758	936.3	6.634	869.5	-7.1%	6.808 ± 0.005	263.2 ± 0.8	6.808 ± 0.002	261.1 ± 0.8	-0.8%	
8	6.660	443.4	6.536	417.2	-5.9%	6.252 ± 0.002	129.7 ± 0.3	6.233 ± 0.006	125.7 ± 0.6	-3.1%	
16	6.612	221.4	6.554	218.1	-1.5%	5.868 ± 0.004	78.7 ± 0.2	5.844 ± 0.001	76.0 ± 0.1	-3.4%	
32	6.649	150.2	6.588	142.0	-5.4%	5.513 ± 0.004	51.4 ± 0.1	5.459 ± 0.006	48.5 ± 0.3	-5.6%	
64	6.654	113.7	6.619	113.0	-0.6%	4.895 ± 0.002	25.4 ± 0.1	4.769 ± 0.043	22.4 ± 0.6	-12.0%	
128	6.649	104.9	6.618	99.6	-5.1%	3.300 ± 0.004	5.8 ± 0.0	3.046 ± 0.055	4.8 ± 0.1	-17.3%	

Variance decomposition. Defining $\mu_F(b) := \mathbb{E}_{\mathcal{K}_b}[F(X_n)]$ and $\sigma_F^2(b) := \text{Var}_{\mathcal{K}_b}[F(X_n)]$ for any statistic $F(X_n)$, the law of total variance gives

$$\text{Var}_{B \sim \pi_s, X \sim \mathcal{K}_B}[F(X_n)] = \underbrace{\mathbb{E}_{B \sim \pi_s}[\sigma_F^2(B)]}_{\text{categorical variability at fixed events}} + \underbrace{\text{Var}_{B \sim \pi_s}[\mu_F(B)]}_{\text{event-pattern variability}}. \quad (13)$$

By Proposition 4, μ_F and σ_F^2 are sampler-independent functions; only π_s depends on the sampler. Proposition 2 controls the count-driven part of π_s . Conditional on S_i^{tot} , SHS forces $J_i \in \{\lfloor S_i^{\text{tot}} \rfloor, \lceil S_i^{\text{tot}} \rceil\}$ with $\text{Var}(J_i \mid S_i^{\text{tot}}) \leq 1/4$, whereas the standard sampler has $\text{Var}(J_i \mid \{p_{ik}\}_{k=0}^{n-1}) = \sum_k p_{ik}(1 - p_{ik})$, approaching the Poisson limit S_i^{tot} as the grid is refined. Under some L_C -Lipschitz sensitivity condition on the count-level conditional mean, this bound propagates to the count-driven component of $\text{Var}_B[\mu_F(B)]$ as $L_C^2 \sum_i \text{Var}(J_i \mid S_i^{\text{tot}}) \leq L_C^2 N/4$ (see Appendix C.3 for the precise statement and derivation). The conditional categorical-variability function $\sigma_F^2(b)$ depends only on the destination kernels $q_{i,k}$, which SHS does not modify. Thus, in this fixed-event view, the only sampler-dependent object is π_s .

Empirical predictions. Two consequences are tested in Section 4. Jump-count variance separates as predicted, with SHS capped at $1/4$ uniformly while the standard sampler grows toward its Poisson limit (Section 4.2). And since only the event-pattern term changes, whatever diversity SHS removes should be path variance rather than destination variance, which Section 4.1 confirms against real text and against the generating model’s own likelihood.

4 Experiments

4.1 Text Generation

We evaluate SHS on four uniform-noise generative language models that span both discrete diffusion (UDLM [Schiff et al., 2025], GIDD [von Rütte et al., 2025], and a 3B-parameter model [von Rütte et al., 2026]) and discrete flow matching (DFM) [Gat et al., 2024], with parameter counts ranging from $\sim 110\text{M}$ to $\sim 3\text{B}$. This selection tests whether the variance-reduction benefit generalizes across model families, training objectives, and scale. In all cases, we keep the trained model (rate/velocity predictor) *fixed* and swap only the inference procedure between the standard sampler and SHS.

Protocol. We compare SHS against the standard sampler of Section 2 at identical NFE, the number of discretization steps n ; one step is one forward pass over all positions, so SHS adds no network evaluations. Generation quality is measured by generative perplexity (Gen. PPL), the exponentiated mean token negative log-likelihood that a fixed GPT2-large evaluator assigns to the generated text [Schiff et al., 2025]; lexical diversity by Distinct- n and Self-BLEU. Each configuration uses 1024 sentences, averaged over 5 seeds for UDLM and GIDD and 3 for DFM; the 3B model uses a single

Table 2: **Lexical diversity under NFE budgets** (Distinct- n higher is more diverse, Self-BLEU lower is more diverse). Single seed, 1024 samples per cell; metric definitions in Appendix E.

NFE	UDLM								GIDD							
	Standard				SHS				Standard				SHS			
	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$
4	0.198	0.758	0.977	0.0000	0.192	0.752	0.975	0.0000	0.251	0.820	0.991	0.0000	0.245	0.814	0.990	0.0000
8	0.188	0.734	0.964	0.0001	0.181	0.730	0.965	0.0000	0.205	0.745	0.972	0.0000	0.195	0.735	0.971	0.0000
16	0.182	0.719	0.955	0.0001	0.170	0.704	0.951	0.0002	0.180	0.702	0.958	0.0001	0.173	0.692	0.956	0.0001
32	0.177	0.702	0.942	0.0003	0.168	0.695	0.942	0.0002	0.170	0.681	0.950	0.0001	0.159	0.664	0.945	0.0001
64	0.174	0.696	0.937	0.0003	0.170	0.699	0.943	0.0002	0.168	0.675	0.946	0.0001	0.156	0.660	0.943	0.0001
128	0.178	0.700	0.938	0.0003	0.167	0.686	0.935	0.0004	0.168	0.676	0.946	0.0001	0.152	0.654	0.941	0.0001
256	0.176	0.697	0.936	0.0002	0.169	0.691	0.935	0.0003	0.168	0.675	0.945	0.0001	0.156	0.657	0.940	0.0002
512	0.173	0.691	0.932	0.0003	0.166	0.688	0.934	0.0002	0.169	0.679	0.946	0.0001	0.151	0.653	0.939	0.0001
1024	0.175	0.695	0.935	0.0003	0.164	0.682	0.930	0.0003	-	-	-	-	-	-	-	-

NFE	3B uniform diffusion								DFM							
	Standard				SHS				Standard				SHS			
	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$	D-1 $\uparrow$	D-2 $\uparrow$	D-3 $\uparrow$	S-BLEU $\downarrow$
4	0.134	0.676	0.968	0.0000	0.120	0.663	0.968	0.0000	0.055	0.439	0.879	0.0004	0.055	0.436	0.877	0.0005
8	0.102	0.591	0.931	0.0001	0.089	0.579	0.929	0.0001	0.032	0.306	0.781	0.0018	0.031	0.301	0.776	0.0017
16	0.088	0.538	0.896	0.0002	0.079	0.535	0.898	0.0003	0.020	0.220	0.686	0.0047	0.019	0.213	0.676	0.0051
32	0.085	0.522	0.880	0.0003	0.076	0.515	0.881	0.0003	0.013	0.156	0.579	0.0116	0.012	0.147	0.560	0.0138
64	0.083	0.510	0.869	0.0005	0.078	0.510	0.873	0.0004	0.009	0.104	0.432	0.0200	0.008	0.099	0.410	0.0221
128	0.082	0.506	0.863	0.0004	0.076	0.501	0.867	0.0005	0.009	0.075	0.265	0.0236	0.008	0.069	0.237	0.0228

seed. The maximum NFE is 1024 for UDLM, 512 for GIDD, and 128 for DFM and the 3B model. For DFM, the model’s local rate λ_{ik} is converted into a per-step change mass with the Euler form $p_{ik} = \min(h\lambda_{ik}, 1)$; Appendix F ablates the frozen-rate form $p_{ik} = 1 - \exp(-h\lambda_{ik})$. Checkpoints, generation lengths and metric definitions are in Appendix E, and matched samples from both samplers in Appendix I.

Results. SHS lowers Gen. PPL at every (model, NFE) budget tested (Table 1), by up to 26% on GIDD and 17% on DFM. Adding SHS on top of nucleus sampling or temperature lowers Gen. PPL further at every setting tested, whereas the adaptive (top-1) scheduling of von Rütte et al. [2026], which also decides when to edit, is $2\text{--}5\times$ worse than the standard sampler (Appendix E.2).

The diversity SHS removes is excess that the standard sampler produces over the real distribution. Table 2 shows SHS slightly below the standard sampler on Distinct- n ; against length-matched real text, the standard sampler is *more* diverse than the corpus it was trained on and SHS moves back toward it, and 88–90% of the lost Distinct-1 is token types absent from the corpus. Fidelity rises at the same time: likelihood under the generating model improves in all seven cells measured, and MAUVE on GIDD rises from 0.381 to 0.536 at NFE=128 (Appendix E.3).

4.2 Jump-Count Variance Diagnostics

We directly verify the variance prediction in Proposition 2. For each generated sample, position, and NFE budget, we record the realized jump count J_i and the cumulative mass $S_i^{\text{tot}} = \sum_k p_{ik}$. Since the theoretical $1/4$ bound is conditional on fixed S_i^{tot} , we estimate conditional count variance by binning positions according to S_i^{tot} and averaging the within-bin variance of J_i .

Table 3: Jump-count variance diagnostics on UDLM. We bin positions by S_i^{tot} (bin width 0.5) and report the sample-weighted average of within-bin variance, an estimator of $\mathbb{E}_{S^{\text{tot}}}[\text{Var}(J_i \mid S_i^{\text{tot}})]$. The standard sampler’s empirical variance grows with NFE as the Poisson-binomial prediction $\sum_k p_{ik}(1 - p_{ik})$ does; SHS remains below the fixed-mass $1/4$ bound (Proposition 2).

NFE	Standard observed	Standard prediction $\sum_k p_{ik}(1 - p_{ik})$	SHS observed	SHS bound
4	0.344	0.405	0.125	0.25
16	1.254	1.642	0.134	0.25
64	2.438	3.096	0.099	0.25
128	3.087	3.845	0.086	0.25

As shown in Table 3, the empirical trend matches the theory: the standard sampler accumulates Poisson-binomial count variance as the grid is refined, whereas SHS sharply concentrates J_i around $\lfloor S_i^{\text{tot}} \rfloor$ or $\lceil S_i^{\text{tot}} \rceil$. This confirms that the proposed intervention removes the intended sampler-induced

count variance rather than merely changing aggregate sample statistics. The concentration is exact: at all 7,864,320 positions logged on UDLM and GIDD, J_i is $\lfloor S_i^{\text{tot}} \rfloor$ or $\lceil S_i^{\text{tot}} \rceil$. The variance SHS removes thus grows with NFE. The Gen. PPL gain in Table 1 need not track this growth, because a uniform-noise model can revise a token written incorrectly at an earlier step and the share of a count deviation that this self-correction absorbs varies by model: the gain grows with NFE on GIDD and DFM and stays within 4.2–7.9% on UDLM and 0.6–7.1% on the 3B model. Drawing θ_i independently per position is what matters: sharing one θ across positions at the same edit budget raises UDLM Gen. PPL at NFE=128 from 174.2 to 213.5, while redrawing it at each unit interval leaves it unchanged (173.7; Appendix E.4).

4.3 Under- and Over-Editing as Count-Deviation Failures

We next test whether deviations from the model-implied cumulative mass are associated with local quality degradation. For each token position, define the count deviation

$$\Delta J_i := J_i - \text{round}(S_i^{\text{tot}}),$$

where $\text{round}(S_i^{\text{tot}})$ is the nearest integer to the cumulative mass $S_i^{\text{tot}} = \sum_k p_{ik}$. We then bin tokens by ΔJ_i and measure token-level evaluator NLL.

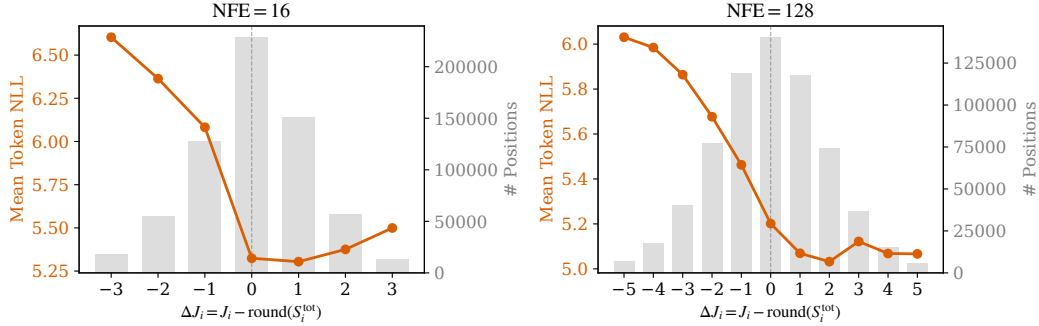


Figure 2: Token-wise NLL versus count deviation $\Delta J_i = J_i - \text{round}(S_i^{\text{tot}})$ on UDLM (Standard sampler) at NFE=16 (left) and NFE=128 (right). Orange line: mean token NLL per deviation bin (left axis). Gray bars: position count per bin (right axis).

Figure 2 shows that token NLL is lowest near $\Delta J_i = 0$ and rises steeply as the realized count falls below the model-implied cumulative mass. The degradation is far more pronounced on the under-edit side ($\Delta J_i < 0$) than on the over-edit side ($\Delta J_i > 0$) at matched $|\Delta J_i|$, and persists within every S_i^{tot} stratum at both NFE budgets (Appendix E.5). SHS confines ΔJ_i to $\{-1, 0, 1\}$ and thus cuts off both tails.

5 Related Work

Sampling rules. A growing line of work improves discrete diffusion/flow samplers by modifying the inference procedure itself. The most lightweight route is destination-distribution shaping via temperature, top- k , or nucleus sampling, which reshapes the destination kernel $q_{i,k}$ at each step [Fan et al., 2018, Holtzman et al., 2020]. JYS optimizes the timestep allocation under a fixed sampling budget [Park et al., 2025]. SCUD conditions discrete diffusion models on the jump schedule, and Neural CTMC explicitly separates the reverse process into an exit-rate component for when to jump and a jump distribution for where to jump [Amin et al., 2025, Li et al., 2026]. Predictor-corrector methods, informed correctors, and Ψ -samplers add or modify correction dynamics to improve sampling [Zhao et al., 2025, Deschenaux et al., 2026]. Planner-based samplers, including DDPD and P2, modify which positions are updated [Liu et al., 2025, Peng et al., 2025]; related first-hitting and token-ordering analyses exploit the order/time structure of masked diffusion [Zheng et al., 2025, Kim et al., 2025]. Few-step and scaling-oriented methods such as FS-DFM and Duo change the training problem, introduce step-budget conditioning, or distill a new sampler [Monsefi et al., 2026,

Sahoo et al., 2025]. Deterministic Discrete Denoising derandomizes the reverse Markov chain by replacing the stochastic denoising transition itself [Suzuki et al., 2025]. In contrast, SHS preserves the learned change masses and destination kernels and modifies only the temporal coupling of the binary stay-vs.-replace events through a single per-position phase, with $O(N)$ overhead and no extra forward passes, hyperparameters, or retraining.

Variance reduction in event simulation. Classical exact simulation of non-homogeneous Poisson processes uses thinning [Lewis and Shedler, 1979, Ogata, 1981], and CTMC inference often uses uniformization to convert continuous-time jump processes into conditionally discrete transition problems [Rao and Teh, 2013, van Dijk et al., 2018]. On the variance-reduction side, systematic and stratified resampling reduce the variance of Monte Carlo resampling relative to independent multinomial sampling in sequential Monte Carlo [Douc et al., 2005], and randomized rounding converts a real-valued mass into an integer count while preserving its expectation [Raghavan and Thompson, 1987, Owen, 2013].

6 Conclusion

We diagnosed a fundamental instability in step-based simulation of CTMC/DTMC discrete diffusion and flow models with a (stay vs. replace) decomposition: the Poisson-binomial jump-count variance grows with NFE toward the cumulative mass $\bar{S}$, inducing under-edit and over-edit failure modes. To address this, we proposed *Systematic Hazard Sampling* (SHS), a drop-in rule that bounds this variance below $1/4$ uniformly in NFE through a single random phase per position, while preserving the per-step event marginals and destination kernel of the standard sampler, without retraining, additional hyperparameters, or extra computation. We confirm this empirically on diverse uniform-noise backbones spanning $\sim 110\text{M}$ to $\sim 3\text{B}$ parameters and covering both diffusion- and flow-based families (UDLM, GIDD, DFM, and a 3B uniform diffusion model): SHS consistently improves generation quality, as reflected in Gen. PPL, and composes with inference-time methods that act on the destination distribution. We hope that SHS contributes to building more robust discrete diffusion and flow models, and that it encourages further research on these models as a promising direction for non-autoregressive generation.

Limitations. SHS is an inference-time variance reduction integrator rather than an exact simulator of the underlying state-dependent CTMC/DTMC process: the per-step change masses $p_{ik}(x)$ are state-dependent, so the realized mass S_i^{tot} is itself random and its distribution may differ between the two samplers, and Propositions 1–2 are accordingly fixed-mass and fixed-event statements. SHS also preserves the expected jump count and reduces only its variance; since the quality cost of count deviation lies on the under-edit side (Section 4.3), biasing the jump count itself is a separate design choice that we do not explore. Our experiments cover uniform-noise discrete diffusion language models, where multi-edit dynamics make jump-count variance most consequential, with a single seed for the 3B model; larger code benchmarks remain future work.

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References

- Alan Nawzad Amin, Nate Gruver, and Andrew Gordon Wilson. Why masking diffusion works: Condition on the jump schedule for improved discrete diffusion. In *NeurIPS*, 2025.
- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *NeurIPS*, 2021.
- Andrew Campbell, Joe Benton, Valentin De Bortoli, Tom Rainforth, George Deligiannidis, and Arnaud Doucet. A continuous time framework for discrete denoising models. In *NeurIPS*, 2022.
- Justin Deschenaux, Caglar Gulcehre, and Subham Sekhar Sahoo. The diffusion duality, chapter II: Ψ -samplers and efficient curriculum. In *ICLR*, 2026.
- Randal Douc, Olivier Cappé, and Eric Moulines. Comparison of resampling schemes for particle filtering. In *ISPA*, 2005.
- Angela Fan, Mike Lewis, and Yann Dauphin. Hierarchical neural story generation. In *ACL*, 2018.
- Itai Gat, Tal Remez, Neta Shaul, Felix Kreuk, Ricky TQ Chen, Gabriel Synnaeve, Yossi Adi, and Yaron Lipman. Discrete flow matching. In *NeurIPS*, 2024.
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *ICLR*, 2020.
- Emiel Hooeboom, Didrik Nielsen, Priyank Jaini, Patrick Forré, and Max Welling. Argmax flows and multinomial diffusion: Learning categorical distributions. In *NeurIPS*, 2021.
- Jaeyeon Kim, Kulin Shah, Vasilis Kontonis, Sham Kakade, and Sitan Chen. Train for the worst, plan for the best: Understanding token ordering in masked diffusions. In *ICML*, 2025.
- P. A. W. Lewis and Gerald S. Shedler. Simulation of nonhomogeneous Poisson processes by thinning. *Naval Research Logistics Quarterly*, 26(3):403–413, 1979.
- Jingyuan Li, Xiaoyi Jiang, Fukang Wen, Wei Liu, Renqian Luo, Yi Zhu, Zuoqiang Shi, and Pipi Hu. Neural continuous-time Markov chain: Discrete diffusion via decoupled jump timing and direction. *arXiv preprint arXiv:2604.15694*, 2026.
- Sulin Liu, Juno Nam, Andrew Campbell, Hannes Stärk, Yilun Xu, Tommi Jaakkola, and Rafael Gómez-Bombarelli. Think while you generate: Discrete diffusion with planned denoising. In *ICLR*, 2025.
- Aaron Lou, Chenlin Meng, and Stefano Ermon. Discrete diffusion modeling by estimating the ratios of the data distribution. In *ICML*, 2024.
- Amin Karimi Monsefi, Nikhil Bhendawade, Manuel Rafael Ciosici, Dominic Culver, Yizhe Zhang, and Irina Belousova. FS-DFM: Fast and accurate long text generation with few-step diffusion language models. In *ICLR*, 2026.
- J. R. Norris. *Markov Chains*. Cambridge University Press, 1997.
- Yoshihiko Ogata. On Lewis’ simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1):23–31, 1981.
- Jingyang Ou, Shen Nie, Kaiwen Xue, Fengqi Zhu, Jiacheng Sun, Zhenguo Li, and Chongxuan Li. Your absorbing discrete diffusion secretly models the conditional distributions of clean data. In *ICLR*, 2025.

- Art B. Owen. *Monte Carlo Theory, Methods and Examples*. 2013.
- Yong-Hyun Park, Chieh-Hsin Lai, Satoshi Hayakawa, Yuhta Takida, and Yuki Mitsufuji. Jump your steps: Optimizing sampling schedule of discrete diffusion models. In *ICLR*, 2025.
- Fred Zhangzhi Peng, Zachary Bezemek, Sawan Patel, Jarrid Rector-Brooks, Sherwood Yao, Avishek Joey Bose, Alexander Tong, and Pranam Chatterjee. Path planning for masked diffusion model sampling. *arXiv preprint arXiv:2502.03540*, 2025.
- Prabhakar Raghavan and Clark D. Thompson. Randomized rounding: A technique for provably good algorithms and algorithmic proofs. *Combinatorica*, 7(4):365–374, 1987.
- Vinayak Rao and Yee Whye Teh. Fast MCMC sampling for Markov jump processes and extensions. *JMLR*, 14(103):3295–3320, 2013.
- Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *NeurIPS*, 2024.
- Subham Sekhar Sahoo, Justin Deschenaux, Aaron Gokaslan, Guanghan Wang, Justin T Chiu, and Volodymyr Kuleshov. The diffusion duality. In *ICML*, 2025.
- Subham Sekhar Sahoo, Jean-Marie Lemerrier, Zhihan Yang, Justin Deschenaux, Jingyu Liu, John Thickstun, and Ante Jukic. Scaling beyond masked diffusion language models. *arXiv preprint arXiv:2602.15014*, 2026.
- Yair Schiff, Subham Sekhar Sahoo, Hao Phung, Guanghan Wang, Sam Boshar, Hugo Dalla-Torre, Bernardo P de Almeida, Alexander Rush, Thomas Pierrot, and Volodymyr Kuleshov. Simple guidance mechanisms for discrete diffusion models. In *ICLR*, 2025.
- Yair Schiff, Omer Belhasin, Roy Uziel, Guanghan Wang, Marianne Arriola, Gilad Turok, Michael Elad, and Volodymyr Kuleshov. Learn from your mistakes: Self-correcting masked diffusion models. *arXiv preprint arXiv:2602.11590*, 2026.
- Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, and Michalis K. Titsias. Simplified and generalized masked diffusion for discrete data. In *NeurIPS*, 2024.
- Hideyuki Suzuki, Wataru Kurebayashi, and Hiroshi Yamashita. Deterministic discrete denoising. *arXiv preprint arXiv:2509.20896*, 2025.
- Nico M. van Dijk, Sebastiaan P. J. van Brummelen, and Richard J. Boucherie. Uniformization: Basics, extensions and applications. *Performance Evaluation*, 118:8–32, 2018.
- Dimitri von Rütte, Janis Fluri, Yuhui Ding, Antonio Orvieto, Bernhard Schölkopf, and Thomas Hofmann. Generalized interpolating discrete diffusion. In *ICML*, 2025.
- Dimitri von Rütte, Janis Fluri, Omead Pooladzandi, Bernhard Schölkopf, Thomas Hofmann, and Antonio Orvieto. Scaling behavior of discrete diffusion language models. In *ICLR*, 2026.
- Guanghan Wang, Yair Schiff, Subham Sekhar Sahoo, and Volodymyr Kuleshov. Remasking discrete diffusion models with inference-time scaling. In *NeurIPS*, 2025.
- Yixiu Zhao, Jiaxin Shi, Feng Chen, Shaul Druckmann, Lester Mackey, and Scott Linderman. Informed correctors for discrete diffusion models. In *NeurIPS*, 2025.
- Kaiwen Zheng, Yongxin Chen, Hanzi Mao, Ming-Yu Liu, Jun Zhu, and Qinsheng Zhang. Masked diffusion models are secretly time-agnostic masked models and exploit inaccurate categorical sampling. In *ICLR*, 2025.

A Visual Diagnostics on Quantized MNIST

A.1 Setup

Our goal is to isolate the effect of event-time sampling on a continuous-time discrete generative model. We fix the pretrained predictor and vary only the sampling procedure: (i) the standard sampler, and (ii) our Systematic Hazard Sampling (SHS).

Few-step regime. To stress-test trajectory instability, we vary the number of discretization steps $\text{NFE} \in \{256, 64, 16, 8\}$ while keeping the same model and schedule. Figure 3 reports hazard-space jump locations at $\text{NFE} = 256$; Figure 4 shows generated samples across all four budgets.

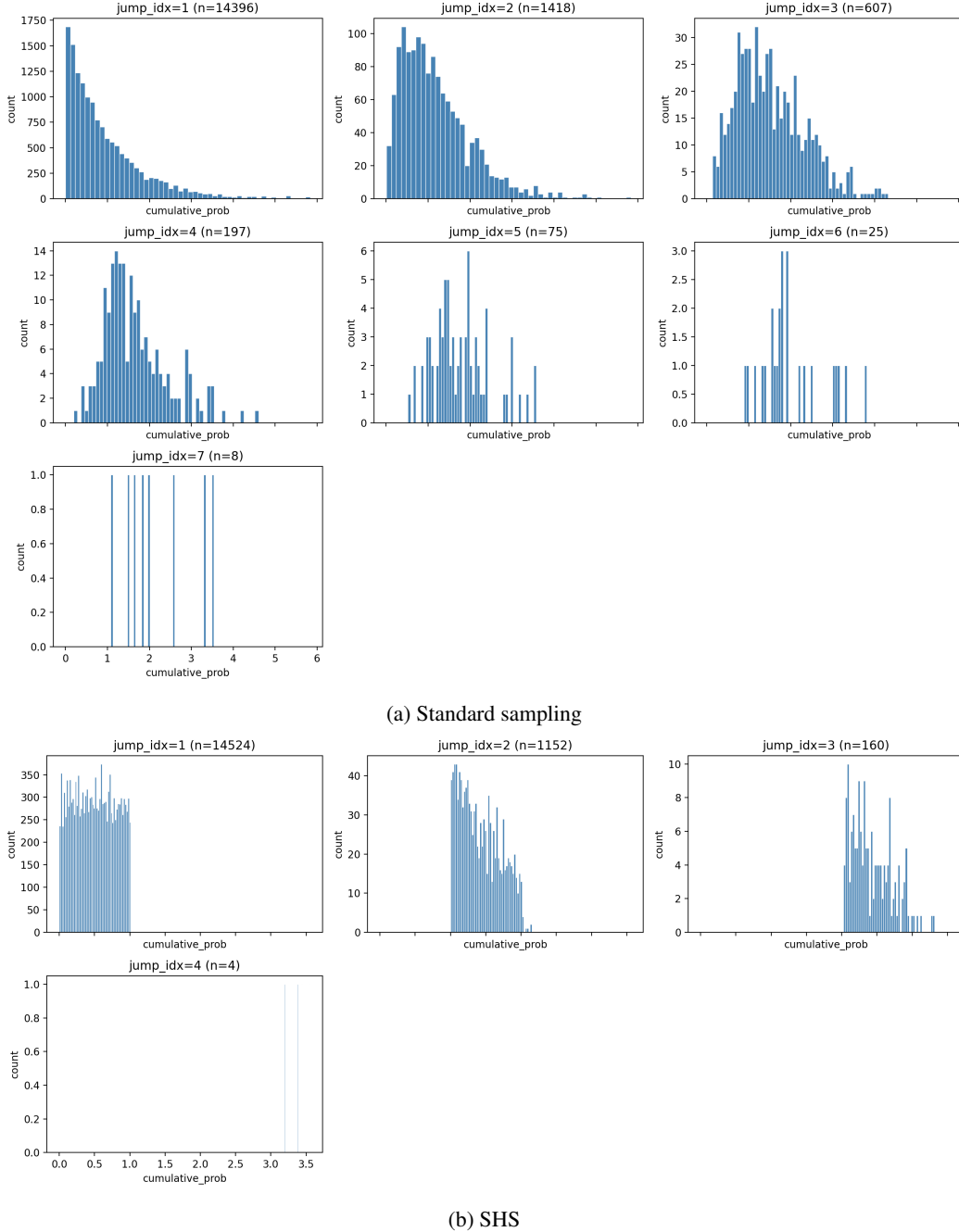


Figure 3: **Hazard-space jump locations** ($\text{NFE} = 256$). Histograms of S_k , the cumulative hazard at the k -th jump. Standard sampling exhibits Erlang/Gamma-shaped variability, while SHS produces systematic, bounded-support locations, demonstrating reduced trajectory randomness.

A.2 Jump Timing Diagnostics in Hazard Space

We first validate whether SHS produces the intended *regularization of event times* without altering the per-jump categorical choice.

For a single sampled trajectory, let $\lambda(t)$ denote the (token-wise) jump rate and define the cumulative hazard

$$S(t) = \int_0^t \lambda(s) ds. \quad (14)$$

We omit the position index i used in the main text since this section examines a single trajectory. We record the hazard-space jump locations $\{S_k\}_{k \geq 1}$, where S_k is the cumulative hazard value at the k -th jump.

Standard baseline: Erlang/Gamma-shaped locations. Under the standard non-homogeneous Poisson process (NHPP) view, inter-event increments in hazard space satisfy $\Delta S_k \sim \text{Exp}(1)$ i.i.d., so the k -th jump location becomes

$$S_k = \sum_{j=1}^k \Delta S_j \sim \text{Gamma}(k, 1), \quad (15)$$

which yields an Erlang-shaped distribution as k increases. Empirically, we observe broad, heavy-tailed variability in early jump locations under standard sampling, consistent with the Poisson-binomial fluctuations analyzed in the main text (Section 2) under fine grids approaching the Poisson limit.

SHS: systematic (bounded-support) jump locations. By construction (Algorithm 1), SHS triggers events at unit-spaced thresholds in cumulative hazard space, constraining the k -th event to a narrow hazard interval. Consequently, the empirical histograms of S_k concentrate with bounded support and avoid the exponential long-tail behavior. This confirms that SHS makes the *timing* of jumps substantially more regular while leaving the per-jump destination sampling unchanged.

A.3 Quantized MNIST Generation: Eliminating Under-Edit

We next visualize the practical effect of trajectory stabilization on a simple discrete generation task. Starting from a noise initialization, we run the sampler for $\text{NFE} \in \{256, 64, 16, 8\}$ discretization steps and compare the resulting samples.

Observation. Standard sampling often exhibits an *under-edit* failure mode in the few-step regime: some positions experience too few jumps, leaving noticeable x_0 -derived noise even late in sampling. In contrast, SHS produces visibly cleaner samples at early stages and substantially reduces residual noise in low-NFE runs. Qualitatively, SHS maintains legible digit structure even at $\text{NFE} = 16$ and $\text{NFE} = 8$, where Standard outputs remain dominated by unresolved noise.

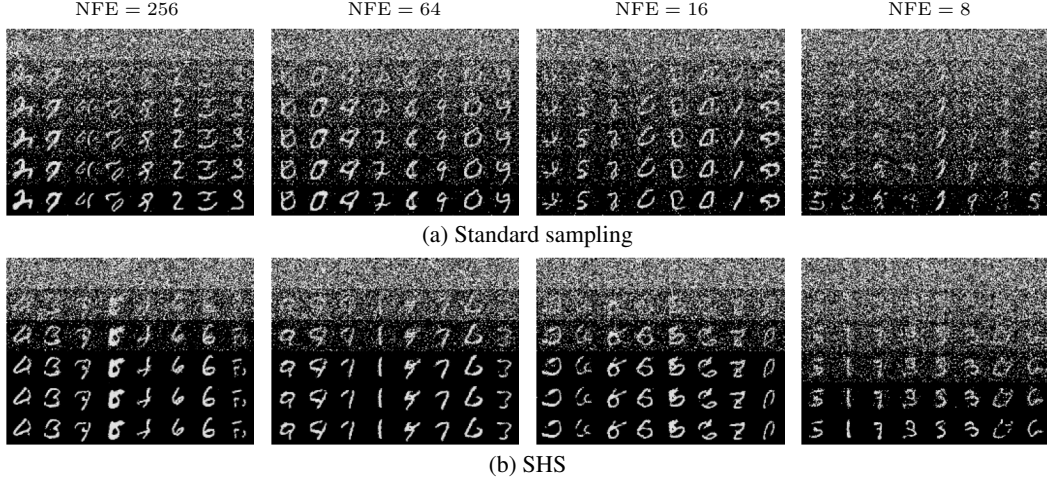


Figure 4: **Quantized MNIST samples under varying step budgets.** Standard sampling (top) vs. SHS (bottom) at $\text{NFE} \in \{256, 64, 16, 8\}$ discretization steps. SHS reduces the under-edit regime and yields cleaner early-stage samples.

B Proofs and Details for Section 3.1

B.1 Proof of Proposition 1 (Jump-Count Unbiasedness)

Fix $S \geq 0$ and draw $\theta \sim \text{Uniform}(0, 1)$. Let $I = \lfloor S \rfloor$ and $f = S - I \in [0, 1)$. Define

$$J = I + \mathbf{1}[\theta < f]. \quad (16)$$

Then

$$\mathbb{E}[J] = I + \mathbb{P}(\theta < f) = I + f = S, \quad (17)$$

since θ is uniform on $[0, 1]$. This proves Proposition 1.

We emphasize again that Proposition 1 is a statement about the integer jump count *conditional on a fixed cumulative mass* S . It does not imply that SHS is distributionally unbiased with respect to the original state-dependent CTMC/DTMC sampler.

B.2 Proof of Proposition 2 (Minimal Variance)

Using the notation from Appendix B.1, we have $J = I + B$ where $B = \mathbf{1}[\theta < f] \sim \text{Bernoulli}(f)$. Since I is deterministic,

$$\text{Var}(J) = \text{Var}(I + B) = \text{Var}(B) = f(1 - f) \leq \frac{1}{4}, \quad (18)$$

where the last inequality follows because the quadratic $f(1 - f)$ attains its maximum at $f = 1/2$.

We now show that this is also the minimum variance among all $\mathbb{Z}_{\geq 0}$ -valued unbiased estimators of S . Let $\tilde{J}$ be any $\mathbb{Z}_{\geq 0}$ -valued random variable with $\mathbb{E}[\tilde{J}] = S$. Write $U := \tilde{J} - I \in \mathbb{Z}$, so that $\mathbb{E}[U] = f$ and $\text{Var}(\tilde{J}) = \mathbb{E}[(\tilde{J} - S)^2] = \mathbb{E}[(U - f)^2]$. For every integer U the identity $(U - f)^2 = U(U - 1) + (1 - 2f)U + f^2$ holds, and taking expectations gives

$$\text{Var}(\tilde{J}) = \mathbb{E}[U(U - 1)] + (1 - 2f)f + f^2 = \mathbb{E}[U(U - 1)] + f(1 - f).$$

Since $U(U - 1) \geq 0$ for every integer U , with equality exactly when $U \in \{0, 1\}$, we obtain $\text{Var}(\tilde{J}) \geq f(1 - f)$, with equality if and only if $\tilde{J} \in \{I, I + 1\}$ almost surely. In that case the unbiasedness constraint $\mathbb{E}[\tilde{J}] = I + f$ forces

$$\mathbb{P}(\tilde{J} = I + 1) = f, \quad \mathbb{P}(\tilde{J} = I) = 1 - f,$$

which is the law of J . Thus the minimum possible variance is $f(1 - f) \leq 1/4$ for any such unbiased estimator, and SHS attains this minimum. This proves Proposition 2.

C Proofs and Details for Section 3.2

C.1 Proof of Proposition 3

Let $S_{i,k} := \sum_{j < k} p_{ij}$ and define the residual threshold $R_{ik} := (\theta_i - S_{i,k}) \bmod 1$. Since $\theta_i \sim \text{Uniform}(0, 1)$ and the modular shift preserves uniformity, R_{ik} is also uniform on $[0, 1)$. SHS triggers an edit at step k exactly when $S_{i,k} + p_{ik}$ crosses a unit-spaced threshold $\theta_i + m$, equivalently when $R_{ik} < p_{ik}$. Hence

$$\mathbb{P}_{\theta_i}(B_{ik}^{\text{SHS}} = 1 \mid \{p_{ij}\}) = \mathbb{P}(R_{ik} < p_{ik}) = p_{ik},$$

identical to the standard Bernoulli(p_{ik}) marginal, proving (11a).

Since $J_i^s := \sum_{k=0}^{n-1} B_{ik}^s$ is the total number of jump events at position i under sampler $s \in \{\text{SHS}, \text{std}\}$, the expectation $\mathbb{E}[J_i^{\text{SHS}} \mid \{p_{ij}\}] = \sum_k p_{ik}$ follows by linearity of expectation, proving (11b).

C.2 Proof of Proposition 4

The event scheduler determines only $B_{ik} \in \{0, 1\}$. Once $B_{ik} = 1$, both the standard sampler and SHS draw the replacement token from $q_{i,k}(\cdot \mid X_{t_k})$. This gives

$$\mathbb{P}^s(X_{t_{k+1}}^i = v \mid B_{ik} = 1, X_{t_k} = x) = q_{i,k}(v \mid x)$$

for both samplers. Note that given an externally specified event pattern b , both samplers start from the same initial law p_0 , keep positions unchanged when $b_{ik} = 0$, and draw from the same destination kernel $q_{i,k}(\cdot \mid X_{t_k})$ when $b_{ik} = 1$. Thus, recursively applying this same update rule over all steps and positions yields the same fixed-event trajectory kernel $\mathcal{K}_b$.

C.3 Count decomposition and Lipschitz propagation

A second application of the law of total variance to the count vector $C(B) = (J_1, \dots, J_N)$ refines the event-pattern variability term in (13):

$$\text{Var}_{B \sim \pi_s}[\mu_F(B)] = \underbrace{\text{Var}_C[\mathbb{E}_{B \sim \pi_s}[\mu_F(B) \mid C]]}_{\text{count-driven}} + \underbrace{\mathbb{E}_C[\text{Var}_{B \sim \pi_s}(\mu_F(B) \mid C)]}_{\text{within-count timing/order}}. \quad (19)$$

Proposition 2 controls the first term through the distribution of the count vector. For sampler s , let

$$\bar{\mu}_{F,s}(c) := \mathbb{E}_{B \sim \pi_s}[\mu_F(B) \mid C = c].$$

If $\bar{\mu}_{F,s}$ admits an L_C -Lipschitz extension from integer count vectors c to $\mathbb{R}_+^N$, then for fixed cumulative mass vector $S = (S_1^{\text{tot}}, \dots, S_N^{\text{tot}})$,

$$\text{Var}_C[\bar{\mu}_{F,s}(C)] \leq \mathbb{E} \left[\left(\bar{\mu}_{F,s}(C) - \bar{\mu}_{F,s}(S) \right)^2 \right] \leq L_C^2 \mathbb{E}[\|C - S\|_2^2]. \quad (20)$$

When the scheduler is count-unbiased for the fixed mass vector S , i.e., $\mathbb{E}[C] = S$, we have $\mathbb{E}[J_i] = S_i^{\text{tot}}$ for each coordinate i . Thus, the right side of (20) is equal to

$$L_C^2 \mathbb{E}[\|C - S\|_2^2] = L_C^2 \sum_{i=1}^N \text{Var}(J_i \mid S_i^{\text{tot}}).$$

For SHS, Proposition 2 gives $\text{Var}(J_i \mid S_i^{\text{tot}}) \leq 1/4$, so the count-driven component is bounded by $L_C^2 N/4$.

C.4 Scope under state-dependent rates

In the full sampler, the change masses $p_{ik} = p_{ik}(X_{t_k})$ are state-dependent. Therefore the observational conditional law $\mathbb{P}^s(X_{0:n} \mid B = b)$ need not equal the fixed-event kernel $\mathcal{K}_b$. This is because, in the observational sampler, the event $B = b$ also encodes information about the visited states through the state-dependent masses $p_{ik}(X_{t_k})$, whereas $\mathcal{K}_b$ treats b as fixed and given. The exact structural statements used in Section 3.2 are fixed-path event-marginal preservation, destination-kernel identity, and fixed-event trajectory-kernel identity. The empirical separation between jump-count variance reduction and corpus-level diversity is evaluated directly in Section 4.1 and Appendix E.3.

D Additional Analysis of Systematic Hazard Sampling

This appendix collects additional properties of SHS and clarifies its scope for CTMC/DTMC-based samplers.

D.1 SHS as a systematic coupling of Bernoulli trials

Consider a fixed sequence of per-step change masses $\{p_k\}_{k=0}^{n-1} \subset [0, 1]$ and define the cumulative sums $S_k = \sum_{j=0}^{k-1} p_j$ (with $S_0 = 0$) and $S_n = \sum_{j=0}^{n-1} p_j$. The standard step-based sampler draws independent indicators $B_k^{\text{iid}} \sim \text{Bernoulli}(p_k)$ and the total number of changes is $J = \sum_k B_k^{\text{iid}}$ (Poisson-binomial).

SHS instead draws a *single* phase $\theta \sim \text{Uniform}(0, 1)$. Write $S_k = I_k + f_k$ with $I_k := \lfloor S_k \rfloor$ and $f_k := S_k - I_k \in [0, 1)$, and note that $S_{k+1} = S_k + p_k$ (so $S_{k+1} = I_{k+1} + f_{k+1}$ with $I_{k+1} := \lfloor S_{k+1} \rfloor$ and $f_{k+1} := S_{k+1} - I_{k+1}$). We define a coupled indicator sequence by

$$B_k^{\text{shs}} = \mathbf{1}[I_k + \mathbf{1}[\theta < f_k] < I_{k+1} + \mathbf{1}[\theta < f_{k+1}]], \quad (21)$$

which is equivalent to triggering a change whenever the cumulative sum crosses the next unit boundary $\theta + m$. Summing (21) over k yields the randomized-rounding form $J^{\text{shs}} = \lfloor S_n \rfloor + \mathbf{1}[\theta < f_n]$ (cf. (9)). We emphasize that this algebraic form serves to make the marginal-preservation and randomized-rounding properties more explicit.

D.2 Event-time systematic placement in cumulative hazard space

For a CTMC with an *exogenous* (state-independent) rate $\lambda(t) \geq 0$, define the continuous cumulative hazard $S(t) = \int_0^t \lambda(\tau) d\tau$. SHS places event boundaries at $\theta, \theta + 1, \theta + 2, \dots$ in hazard space, so the m -th event time is

$$T_m = \inf\{t \in [0, 1] : S(t) \geq \theta + (m - 1)\}. \quad (22)$$

Equivalently, the hazard value at event m is exactly $\theta + (m - 1)$. Thus, SHS guarantees *exactly one* event in each unit-length hazard interval, eliminating the exponential-tail inter-arrival variability of Poisson processes in hazard space (where $\Delta S \sim \text{Exp}(1)$).

On a discrete grid (DTMC kernels or CTMC τ -leaping), S_k increases in steps. If each increment satisfies $p_k \leq p_{\max} \leq 1$, then the boundary-crossing event triggered at step k overshoots its hazard boundary by at most $p_{\max}$. Therefore, SHS still yields tightly controlled event locations in hazard/jump-mass space even under discretization.

Timing of individual edits. SHS moves edits slightly earlier in hazard space and makes their timing less variable. In cumulative-hazard space the m -th edit occurs at $\theta + (m - 1)$, with mean $m - \frac{1}{2}$ and variance $\frac{1}{12}$, against $\text{Gamma}(m, 1)$ with mean m and variance m under the standard sampler. Conditionally on no edit having been triggered yet, the probability of an edit at the next step rises (with $p_1 = 0.8$ and $p_2 = 0.15$, it is $0.15/0.2 = 0.75$ after a step without an edit), but this state is reached with probability 0.2, so the per-step marginal remains $0.2 \times 0.75 = p_2$ (Proposition 3); this coupling across steps is what removes the count variance. On UDL, the mean jump count per position under SHS exceeds the standard sampler's by 2.1/2.7/3.1% at NFE 16/64/128, yet the final-step edit rate falls to 0.43/0.15/0.11 $\times$: SHS edits more often than the standard sampler up to $k/n \approx 0.75$ –0.81 and less often afterwards, independently of n , so edits are redistributed toward the front of the trajectory.

D.3 Scope: inference-time variance reduction vs. exact simulation

The results in Propositions 1 and 2 are unconditional statements about randomized rounding for a *fixed* mass S (or a fixed sequence $\{p_k\}$). In a full generative model, the per-step masses $p_{ik}(x)$ (or rates $\lambda_i(t, x)$) are typically *state-dependent*. Because SHS couples change decisions over time through the phase θ_i , it introduces additional memory beyond the visible state x . Accordingly, SHS should be viewed as an *inference-time variance reduction integrator* for CTMC/DTMC samplers rather than as an exact simulator of the underlying Markov process.

Despite this, SHS leaves the per-change destination kernels $q_{i,k}$ unchanged and dramatically reduces sampler-induced variability in the number and timing of edits. This reduction is especially valuable under uniform-noise initialization, where meaningful generation requires multiple self-correction edits per position and where Poisson-binomial fluctuations can dominate the observed instability.

D.4 Absorbing-state (mask-start) implies single-jump trajectories

Here, we formalize why the jump-count variance discussed in Section 2 is a *multi-edit* phenomenon and is absent under the standard absorbing-state (mask-start) setting.

Let $\mathcal{V}$ be the vocabulary and let $m = [\text{MASK}]$ be a dedicated sentinel token. Define the extended alphabet $\bar{\mathcal{V}} := \mathcal{V} \cup \{m\}$, and consider step-based sampling on a grid $\{t_k\}_{k=0}^n$. A state is $x \in \bar{\mathcal{V}}^N$. At each step k and position i , assume the sampler admits a (stay vs. replace) decomposition as in Section 2, with a change mass $p_{ik}(x) \in [0, 1]$ and a destination distribution $q_{i,k}(\cdot | x) \in \Delta(\mathcal{V})$.

Assumption 5 (Unmask-only (Absorbing) Dynamics). The absorbing-state (mask-start) sampler abides by the following *unmask-only* dynamics: once a position leaves m , it never changes again:

$$p_{ik}(x) = 0 \quad \text{whenever } x_i \in \mathcal{V}, \text{ for any } x \in \bar{\mathcal{V}}^N. \quad (23)$$

Proposition 6 (Single-edit property under absorbing-state dynamics). *Under Assumption 5, each position can undergo at most one edit: for every i , $J_i \in \{0, 1\}$ almost surely.*

Proof. If $J_i = 0$, the statement is obvious. Otherwise, define the (non-trivial) edit indicator and total jump count

$$B_{ik} := 1[x'_i \neq x_i], \quad J_i := \sum_{k=0}^{n-1} B_{ik}, \quad (24)$$

and let k^* be the first step where $B_{ik^*} = 1$. When $x_i = m$, the update at step k draws $B_{ik} \sim \text{Bernoulli}(p_{ik}(x))$ and, if $B_{ik} = 1$, sets $x'_i \sim q_{i,k}(\cdot | x)$ (thus $x'_i \in \mathcal{V}$); otherwise it keeps $x'_i = m$. Hence $x_i^{(k^*)} = m$ and after the update we have $x_i^{(k^*+1)} \in \mathcal{V}$. Then Assumption 5 implies that for all later steps $\ell > k^*$, $p_{i\ell}(x^{(\ell)}) = 0$ and thus $B_{i\ell} = 0$. Thus, only one edit can occur, so $J_i = 1$. $\square$

Proposition 6 has the following implications on the variance. Since $J_i \in \{0, 1\}$, it is a Bernoulli random variable and

$$\text{Var}(J_i) \leq \frac{1}{4}. \quad (25)$$

Moreover, many mask-start implementations explicitly force any remaining [MASK] tokens to be resolved by the terminal time (e.g., by setting $p_{i,n-1}(x) \equiv 1$ for $x_i = m$ at the final step), in which case $J_i = 1$ deterministically and $\text{Var}(J_i) = 0$.

Interpretation (why SHS is mainly needed for uniform-noise start). Under mask-start, the state explicitly reveals whether a position has been generated via the sentinel $m = [\text{MASK}]$, and the unmask-only constraint (23) enforces a single-edit trajectory per position. In contrast, uniform-noise start typically requires multiple self-correction edits per position, where sampler randomness in the number and timing of edits becomes substantial (Section 2), motivating SHS as a minimal-variance event scheduler.

E Experimental Details and Additional Results

E.1 Setup

Checkpoints are kuleshov-group/udlm-lm1b (UDLM, trained on LM1B), dvruette/gidd-base-p_unif-0.2 (GIDD), dvruette/gidd-unif-3b (3B), and the 1.3B DFM checkpoint released with FS-DFM [Monsefi et al., 2026]. Generation length follows each model’s own configuration: 128 tokens (UDLM), 512 (GIDD), 256 (3B), 1024 (DFM); generation is unconditional. Gen. PPL is $\exp(\sum_{m,t} -\log p_{\text{GPT2}}(x_{m,t} | x_{m,<t}) / \sum_m T_m)$ over all generated tokens, scored on the decoded text re-tokenized by GPT-2 (for DFM, on the generated token ids directly); empty samples are excluded. Its absolute value depends on the evaluator, the generation length and the training corpus, so Gen. PPL is compared only between the two samplers within a model. GIDD samples are scored without the self-correction pass of its released pipeline. Entropy is the unigram entropy (nats) of the generated tokens pooled over the 1024 samples; for DFM it is the per-sequence entropy in bits, averaged over samples. Distinct- n is the fraction of unique n -grams over whitespace tokens pooled over samples; Self-BLEU is BLEU-4 of 500 randomly drawn samples against 50 random references with a fixed RNG. Special tokens are retained or stripped according to each model’s own decoding, so diversity is compared only between the two samplers within a model.

E.2 Additional Gen. PPL results

On UDLM, SHS at NFE=128 (175.7) is below the standard sampler at NFE=1024 (181.6), so the gain on this model exceeds that of an eightfold increase in NFE. SHS decides *when* a position is edited and never *what* it is replaced with. Adaptive scheduling also decides when to edit, but keeps only the ranking of the change probabilities and discards their values, which SHS matches in expectation; it raises entropy by about 1.8 nats and Gen. PPL by $2\text{--}5\times$ (Table 4), producing high-entropy text rather than sharper text. Nucleus sampling and temperature reshape the destination distribution instead: both are applied to the destination distribution after the edit decision, so only the replacement token changes, and both compose with SHS at every setting (Table 5). At matched entropy, SHS alone lies on the temperature frontier (Gen. PPL 84.8 at entropy 7.234, against ≈ 85.4 for temperature interpolated to the same entropy; bootstrap difference -0.7 , 95% CI $[-3.4, +2.6]$) without modifying $q_{i,k}$, and adding SHS on top of temperature lowers Gen. PPL at every τ .

Table 4: Same-axis comparison on the 3B model (single seed). Entropy / Gen. PPL.

	NFE	Standard	Adaptive (top-1)	SHS
	4	6.758 / 936.3	8.248 / 1995.0	6.634 / 869.5
	8	6.660 / 443.4	8.384 / 1483.8	6.536 / 417.2
	16	6.612 / 221.4	8.459 / 1094.8	6.554 / 218.1
	32	6.649 / 150.2	8.454 / 783.7	6.588 / 142.0
	64	6.654 / 113.7	8.421 / 591.9	6.619 / 113.0
	128	6.649 / 104.9	8.483 / 574.9	6.618 / 99.6

Table 5: Composition with nucleus sampling and temperature at NFE=128 (mean of 3 seeds). Entropy / Gen. PPL.

Model	Standard	top- p 0.9	top- p 0.9 + SHS
UDLM	6.884 / 190.65	6.866 / 187.45	6.817 / 174.55
GIDD	7.350 / 111.63	7.299 / 97.56	7.206 / 78.23
GIDD, τ		Temperature	Temperature + SHS
1.0		7.350 / 111.63	7.234 / 84.83
0.99		7.301 / 99.89	7.182 / 76.09
0.975		7.218 / 81.91	7.118 / 65.02
0.95		7.081 / 63.34	7.002 / 51.72
0.90		6.853 / 40.57	6.781 / 34.86

E.3 Diversity and fidelity

The standard sampler is more diverse than real text and SHS moves toward real text (Table 6); the reference is OpenWebText cut into blocks matching the average word count of the generated texts, and all metrics use whitespace tokens. For $\text{NFE} \geq 16$, 88–90% of the Distinct-1 reduction is token types absent from the reference corpus; tokens common in the corpus contribute under 2%. Under the generating model’s own per-token ELBO (Monte Carlo over diffusion time, 128 draws for UDLM and 64 for GIDD, seeds 1–3), SHS samples score lower in every cell: UDLM at NFE 16/32/128/512, 4.676 $\rightarrow$ 4.580, 4.459 $\rightarrow$ 4.396, 4.357 $\rightarrow$ 4.265, 4.315 $\rightarrow$ 4.213; GIDD at NFE 16/128/512, 5.151 $\rightarrow$ 4.946, 4.394 $\rightarrow$ 4.195, 4.337 $\rightarrow$ 4.049. MAUVE (GPT2-large features, 102 buckets, 1024 samples per seed against 5000 reference texts: LM1B validation for UDLM, OpenWebText for GIDD) rises on GIDD from 0.237 to 0.303 (NFE 32), 0.381 to 0.536 (128) and 0.418 to 0.559 (512); on UDLM both samplers stay below 0.15, too low to separate.

Table 6: GIDD lexical diversity against length-matched real text (std / SHS).

	D-1	D-2	D-3	Self-BLEU-2
Real data	0.148	0.626	0.903	0.062
NFE 16	0.180 / 0.171	0.702 / 0.688	0.958 / 0.955	0.057 / 0.059
NFE 128	0.168 / 0.153	0.677 / 0.656	0.946 / 0.941	0.053 / 0.056
NFE 512	0.169 / 0.153	0.678 / 0.654	0.946 / 0.939	0.052 / 0.054

E.4 Verification under state-dependent sampling and phase ablation

Measured on the masses each position actually accumulated under state-dependent sampling, the invariant $J_i \in \{[S_i^{\text{tot}}], [S_i^{\text{tot}}]\}$ holds at all 7,864,320 logged positions, the conditional variance repeats with the unit period in S_i^{tot} that $f(1 - f)$ predicts (measured period 0.966–1.004, NFE 8–1024), and its magnitude is $2.8\times$ (NFE 4) to $48.8\times$ (NFE 1024) below the standard sampler’s. Table 7 ablates the phase on UDLM at NFE=128 with 3 seeds and paired tests: the shared setting draws one θ per sequence for all positions, the redrawn setting replaces a position’s θ after each of its edits. The shared and per-position settings agree in $\mathbb{E}[S^{\text{tot}}]$ to within 0.16%, so the 39-point gap comes from the coupling across positions rather than from the edit budget. The law of θ and the unit threshold spacing are not free choices: Propositions 1–2 force both.

Table 7: Phase ablation on UDLM, NFE=128, 3 seeds.

Phase θ	Gen. PPL	paired p vs. per-position	$\mathbb{E}[S^{\text{tot}}]$
Independent per position (SHS)	174.248	–	5.0789
Shared across positions	213.477	0.0039	5.0868
Redrawn at each unit interval	173.711	0.799	–

E.5 Count deviation stratified by cumulative mass

Token NLL in Figure 2 is the GPT2-large NLL mapped to each model position by character offsets. Stratifying by S_i^{tot} removes 81–84% of the raw excess NLL of positions with $\Delta J_i \neq 0$ over positions with $\Delta J_i = 0$ ($0.287 \rightarrow 0.055$ at NFE 16, $0.127 \rightarrow 0.020$ at NFE 128) and leaves the under-edit penalty significant in all 12 strata; the over-edit side is significantly negative in 10 of 12 (Table 8).

Table 8: Excess token NLL (nats) relative to $\Delta J_i = 0$ within each S_i^{tot} stratum (width 1), UDLM standard sampler, seeds 1–5. Brackets: 95% CI, sentence-level cluster bootstrap.

S_i^{tot}	NFE = 16		NFE = 128	
	under-edit ($\Delta J_i \leq -1$)	over-edit ($\Delta J_i \geq +1$)	under-edit ($\Delta J_i \leq -1$)	over-edit ($\Delta J_i \geq +1$)
[1, 2)	+1.991 [+1.910, +2.070]	−0.259 [−0.312, −0.208]	+0.716 [+0.509, +0.927]	−0.350 [−0.466, −0.238]
[2, 3)	+0.946 [+0.899, +0.995]	−0.445 [−0.478, −0.414]	+0.608 [+0.512, +0.708]	−0.366 [−0.442, −0.293]
[3, 4)	+0.397 [+0.359, +0.436]	−0.155 [−0.189, −0.119]	+0.543 [+0.481, +0.602]	−0.286 [−0.337, −0.237]
[4, 5)	+0.281 [+0.236, +0.324]	−0.051 [−0.099, −0.003]	+0.307 [+0.261, +0.353]	−0.182 [−0.221, −0.144]
[5, 6)	+0.375 [+0.292, +0.457]	−0.110 [−0.206, −0.003]	+0.177 [+0.132, +0.220]	−0.097 [−0.141, −0.053]
[6, ∞)	+0.248 [+0.135, +0.358]	+0.061 [−0.122, +0.243]	+0.110 [+0.066, +0.157]	−0.045 [−0.094, +0.004]

F Ablation: DFM Change-Mass Discretization

DFM samplers require converting a local model output λ_{ik} into a valid per-step change mass $p_{ik} \in [0, 1]$. We ablate the choice of discretization, comparing two forms:

$$p_{ik}^{\text{Euler}} = \min(h\lambda_{ik}, 1), \quad p_{ik}^{\text{exp}} = 1 - \exp(-h\lambda_{ik}).$$

The exponential form is the frozen-rate CTMC event probability. The Euler form is the first-order τ -leaping mass used in our main DFM experiments. Since $1 - \exp(-x) \leq x$, the exponential form yields a smaller effective cumulative mass for positive x relative to the model’s intended flow λh .

Table 9: DFM generation quality (Gen. PPL, lower is better) under the two change-mass discretizations, standard sampler, 320 samples, single seed; $\pm$ is the standard deviation over samples. The percentage column reports the relative reduction of p^{Euler} over p^{exp} .

NFE	$p^{\text{exp}} = 1 - \exp(-h\lambda)$	$p^{\text{Euler}} = \min(h\lambda, 1)$	Δ
2	1150.5 $\pm$ 157.9	759.2 $\pm$ 94.5	−34.0%
4	372.7 $\pm$ 55.8	286.5 $\pm$ 39.5	−23.1%
8	158.4 $\pm$ 21.8	141.0 $\pm$ 17.3	−11.0%
16	86.3 $\pm$ 12.1	83.9 $\pm$ 10.3	−2.8%
32	52.3 $\pm$ 10.3	53.4 $\pm$ 7.3	+2.1%
64	29.5 $\pm$ 10.3	28.7 $\pm$ 10.6	−2.7%

The Gen. PPL gap in Table 9 at small NFE motivates using the Euler change mass for the DFM experiments reported in the main text.

G Broader Impact

SHS is a training-free, hyperparameter-free inference principle that can be dropped into any CTMC/DTMC sampler admitting a stay-vs.-replace decomposition, at virtually no computational cost

and with no extra forward passes. By achieving the minimum possible jump-count variance ($\leq 1/4$) for any fixed cumulative mass, it directly addresses a sampler-induced failure mode—under- and over-editing—identified empirically in our diagnostics. The consistent improvements in generative perplexity across diverse uniform-noise backbones spanning $\sim 110\text{M}$ to $\sim 3\text{B}$ parameters and covering both diffusion- and flow-based families (UDLM, GIDD, DFM, and a 3B uniform diffusion model) indicate that the benefit stems from the sampler rather than from model-specific artifacts.

Since SHS preserves the learned destination kernels and only changes the temporal coupling of stay-vs.-replace events, it does not alter what the underlying model is willing to produce. Its societal risk profile is therefore inherited from the underlying generative models, and the same considerations around misuse of any text generator apply.

H Hardware Specifications

All experiments were conducted on a system with 256 CPU cores, 4 Nvidia RTX L40S (48GB) GPUs, and 1024 GB of RAM. In total, the experiments, evaluations, analyses, and method development required approximately 500 GPU hours.

I Qualitative Examples

Samples are matched by model, NFE and seed, taken in stored order (indices 0–2) without selection, cut at 60 words for display, with per-sample Gen. PPL; in the UDLM pairs, tokens are colored by ΔJ_i (red under-edited, blue over-edited, darker for larger $|\Delta J_i|$).

UDLM, NFE=16, seed 1.

Standard, idx 0 (Gen. PPL 551.0)

[CLS] waterford , mich . , june 21 / prnewswire / - - neverland has returned to purne to be the first of the u . s . ships elizabeth europe . along to bring a lot more food and visitors . velemouth had \$ 1 fighter arena designed at port 1 mount st . , according to susbastle . [...]

Standard, idx 1 (Gen. PPL 96.5)

[CLS] in other words , says rendi . so credit in the world leader has the resources to find urgently needed relief and referrals to other countries and economies . and since haiti was born in latin america , which the united states signed upon by france , its position - - as most influential nation in western nations - [...]

Standard, idx 2 (Gen. PPL 231.0)

[CLS] it was last updated at 01 . 39 gmt on saturday 24 february 2009 . editorial associates . central london with a francis per annum bu £25000 - £24000 per annum + benefits . the post office will provide . you will with direct and marketing remun ba £60 , 000 [...]

SHS, idx 0 (Gen. PPL 403.0)

[CLS] one last of a kind sun - tan post - exhibition and policy conference run by governmentolov , ms bleas vsa and more often will be the leaders of an al qaeda site , tehran . put on madak ' s performance as the director of pakistan ' s " greifi now " book watch network , a mahdi [...]

SHS, idx 1 (Gen. PPL 216.1)

[CLS] irrelevant . yeah u . s . follow or maybe we are going to . to getting autos only go on making it for a budget market because they do . the american healthcare bill more than adequate - it provides no new help for these people who have that problem . more anxious and bothered for the sake [...]

SHS, idx 2 (Gen. PPL 168.7)

[CLS] a whiz on " the only trouble thinking americans " when it comes off the front . back in america ' s ' 70s that turn of the head down 14th street where cashed ' s death was touched was a beginning for both my o3 and our most important lunch audience . but , the " part dr [...]

UDLM, NFE=128, seed 1.

Standard, idx 0 (Gen. PPL 176.5)

[CLS] she was a successful student and advocate for occ and its mission including women and retirees and workers of the city from 1996 to 2004 ' it entrepreneurship ' for the local business community . the web site began operating today shortly before visitors . this weekend , qe 1 . com decided to dispatch writer , ezra eszelan [...]

Standard, idx 1 (Gen. PPL 353.1)

[CLS] circa £33 , 000 . to plan and work in an online video game . stem fisher . a doctor offers geo stories (you have a weekly o £35000 - £4299 per annum + performance allowance or on clearance level . - - higher education education : to work with a u . . [...]

Standard, idx 2 (Gen. PPL 275.9)

[CLS] existing optafar base price requirements may even be reflected in comparable product to improve conditions or demand ; the consumption of dvrs and principally ; the overall release to its product backlog to attract new sales programs and expanding services that will lower gross revenues in other measures by a net one percent : a reduction in service and [...]

SHS, idx 0 (Gen. PPL 272.1)

[CLS] we really had a continuing story during the first full week of 2008 , but these builders and developers entered this inevitable new phase of disfiguring the stock market , ” said wgc analyst john dragonog , looking back now that lorillard has closed 11 ” dealerships ” at an 18 - member equipment shop chain ’ s location [...]

SHS, idx 1 (Gen. PPL 199.6)

[CLS] it was last updated at 11 . 24 bst on thursday 1 may 2009 . liberty is an early education center based on fantastic or lig £35 , 000 to £33 , 799 per annum + 15 % contribution to £20 , 000 £37 , 933 to £44 , 833 pa . base - salary [...]

SHS, idx 2 (Gen. PPL 165.4)

[CLS] spectra channelau ’ s recent fiscal year 2009 gold production when it attributes to the expansion in the dollar market . the quarter used to offset a slight increase in additions by customers and the rate of redflows and in both the prices of low - yielding currencies . over the winter the company expects volumes to be higher [...]

3B model, NFE=128, seed 0 (corpus Gen. PPL: Standard 104.9, SHS 99.6).

Standard, idx 0 (Gen. PPL 88.0)

of content confirmed expanded the club , and the influx of interest and information requests has caused a significant content shortage on the club ’ s website . With even talent basic , the .COM is forced to develop an array of other formats for the community . Marko notes , ” We are grateful for your continued support , but [...]

Standard, idx 1 (Gen. PPL 200.1)

stationery x . 5 / wide / 1 x . 5 / high / 4 x . e / warmth Noonan 2 x . 5 / high . 5 x . 75 / depth / 2 x . 5 / deep / Important support us please click / If would you like to reply us ? / Click here . [...]

Standard, idx 2 (Gen. PPL 190.3)

joy comes out of I , and I look forward to returning for more . ” soon / importance i was really charged by your , S and your day what ’ s calling the common genre , and like the direction of Jesus songs think it is good bit mysteries , but I will be more watching [...]

SHS, idx 0 (Gen. PPL 53.5)

had never been alone before and she was nervous on nervous . ” I guess , if you want to take off , I think you can wear go with me . ” Her housemate ’ s tone was concerned . she didn ’ t have any words for herself but nevermind her face . She tried to leave the room walking slowly around [...]

SHS, idx 1 (Gen. PPL 148.3)

as normal on Monday night . Looking should be a real strong race tracks on the promotion crew . Week Revival has been attracting racers in and out since 1981 is the 5th International Trail . It is one of the no one races for Monaco major 's held in Columbus , Indianapolis , Atlanta , Wisconsin , [. . .]

SHS, idx 2 (Gen. PPL 94.9)

player that tends to develop a habit of making good bets , although a player that should adjusted Not will be too willing to protect their bluff and will get in the pot , making other players seem weak . Putting Your / -Game Action : / Now , you all know how , and we all know how . [. . .]

GIDD, NFE=512, seed 1 (corpus Gen. PPL: Standard 103.4, SHS 76.6).

Standard, idx 0 (Gen. PPL 73.0)

had the opportunity to watch Guardians embryonic family again with Dave Bautista as well as Victoria Sumpter and Gwyneth Paltrow. The actors had to keep together their rig while wearing clothes and makeup, he explained down the line. The shoot and movie music will be screened by one or a few fans. [. . .]

Standard, idx 1 (Gen. PPL 229.0)

, lotteries involving war movies, a comedy series with the Hound of Bjermart, a rock band ThreeGo between Fanon poll and O2 legends Hillfritoww, and a huge rival monthly club night, with a running variety of films, an occasional radio station and a campaigner video store. [. . .]

Standard, idx 2 (Gen. PPL 250.6)

through the anime Pursusta, if your not college drama manga. Her ticks? Because you chose to deserves to date with two fisticuffs only to make her crumpas pay. Go for: Jacket, accessories, starting trouble, experiences, if the answer is sensible/ ponytail:MamaLux, a & interview convention [. . .]

SHS, idx 0 (Gen. PPL 76.6)

Others arriving with a special ability. Mr. Reynoso underscored a "thin line between the Marines students with each lousy group of Marines and Army outposts." Photograph by Devin Lee. "He had a bar band hit, made up of simply lined up in with him, whatever the music was," Hamere said. [. . .]

SHS, idx 1 (Gen. PPL 123.5)

"What's the best hope in the future for policies in store?" was Microsoft's reply to the announcement late after the night that Microsoft Store would permanently slapped down Windows Phone developers. Usability normally the most regrettable will help soften the blow, and has been chosen to kick things off. [. . .]

SHS, idx 2 (Gen. PPL 159.8)

soil. Biotechnology is also a key to continue to bring farming plants, though Kussman believes other ways to fish are sustainable. (Photo of one fish with kuyi in aquaponics. (AE image, Jim Stornian and Elisena Allill, FPL) By 2018, the FPL anomalyland farmers will be self-sufficient. [. . .]